Chapter 1

Introduction to Enterprise Resource Planning (ERP)

Learning objectives

- Overview of ERP.
- Business function vs business process.
- Client/Server Technology, 2 Tier, 3 Tier, n Tier.
- Example of ERP – SAP.
- SAP S/4HANA (using Global Bike).

1. Overview of ERP

What is ERP?

- ERP Systems integrates and automates many internal business processes and information systems within the manufacturing, logistics, distribution, accounting, finance and human resource functions of company.

- ERP is an integrated software program which combines all the business processes of different department and function across a company onto a single computer system that can serve particular needs of the different departments, and they can more easily share information and communicate with each other.

What is ERP? (Cont.)

- For example, someone in finance department can use ERP to see if any sale order has been shipped from the warehouse so that he can now confidently plan for working capital management for the next period.
- Common ERP software modules include product planning, purchasing, inventory, distribution, order tracking, finance, accounting, and other similar things.
- The users of ERP systems are employees of the organization at all levels, such as top management, mid-level managers, supervisors, workers, and executives.

What is ERP? (Cont.)

- The goal of ERP is to improve and streamline internal business processes, which typically requires reengineering of current business processes.
- ERP systems provide a mechanism for implementing systems where a high degree of integration between applications is required.

Example of Major ERP modules

- Manufacturing: Some of the functions include engineering, capacity, workflow management, quality control, bills of material, manufacturing process, etc.
- Financials: Accounts payable, accounts receivable, fixed assets, general ledger and cash management, etc.
- Human Resources: Benefits, training, payroll, time and attendance, etc.
- Supply Chain Management: Inventory, supply chain planning, supplier scheduling, order entry, purchasing, etc.

Example of Major ERP modules (Cont.)

- Projects: Costing, billing, activity management, time, expense, etc.
- Customer Relationship Management: sales and marketing, service, commissions, customer contact, call center support, etc.
- Data Warehouse: A module can be accessed by an organization's customers, suppliers, employees, etc.
- Other similar modules.

Benefits of ERP

- Implementing ERP results in significant improvements in the *quality and efficiency* of customer service, production and distribution.
- There is *a significant reduction* in transaction processing costs, hardware, software, etc.
- ERP provides vital information to managers quickly and timely, which helps with *decision-making. (Good Reporting capabilities)*
- Implementing ERP systems breaks down many departmental and functional walls.
- Supports multiple currencies and languages.

Challenges of ERP

- Though the benefits of ERP are many, the costs and risks of implementing it are high.
- Complex software.
- The ERP implementation failure causes losses in revenue, profits, and market share when it affects orders, shipments, inventory, etc.

Causes of ERP failures

- Failed to provide planning, development, and training to ERP users.
- Failed to involve affected employees in the planning and developmental stages of the ERP system.
- Too fast in the conversion process (from the existing system to the new ERP system).
- It is better to implement the ERP system module by module instead of implementing it as a whole and replacing the existing system suddenly.

Development of ERP Systems

SAP – The Intelligent Enterprise

- Market leader in enterprise application software
- One of the world largest independent software manufacturer
- 400,000 satisfied Customers in 190 countries
- SAP enables companies to
  - Streamline processes
  - Use live data
  - Predict customer trends
  - Connect entire businesses

Intelligence will reinvent industries and change business forever

Best-run technologies and solutions deployed end-to-end for your industry.

Figure 1: SAP – The Intelligent Enterprise. Source: Introduction to SAP S4HANA @2023 SAP SE/SAP UCC Magdeburg. All rights Reserved.

Development of ERP Systems (Cont.)

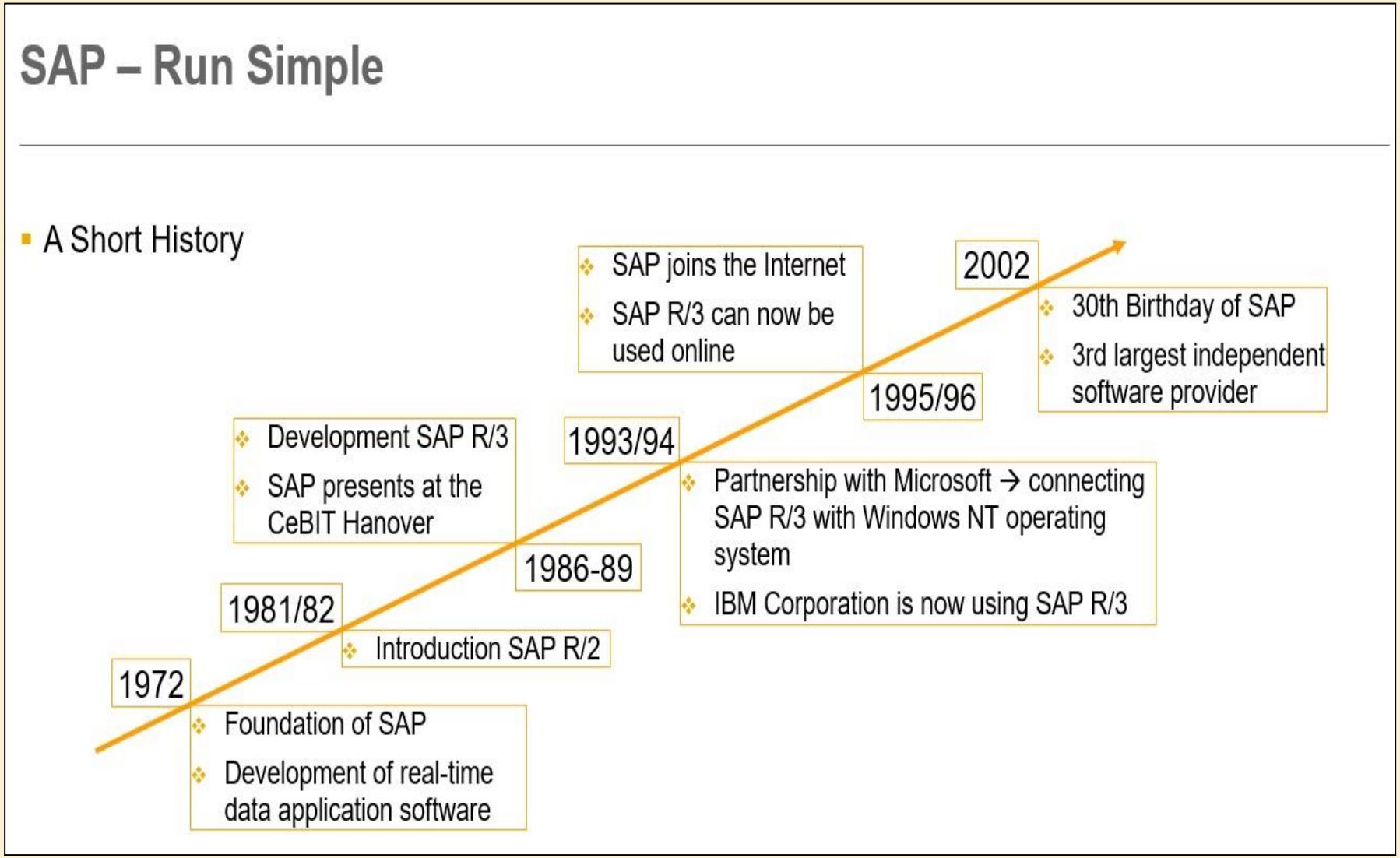


Figure 2: SAP – Run Simple. Source: Introduction to SAP S4HANA @2023 SAP SE/SAP UCC Magdeburg. All rights Reserved.

Development of ERP Systems (Cont.)

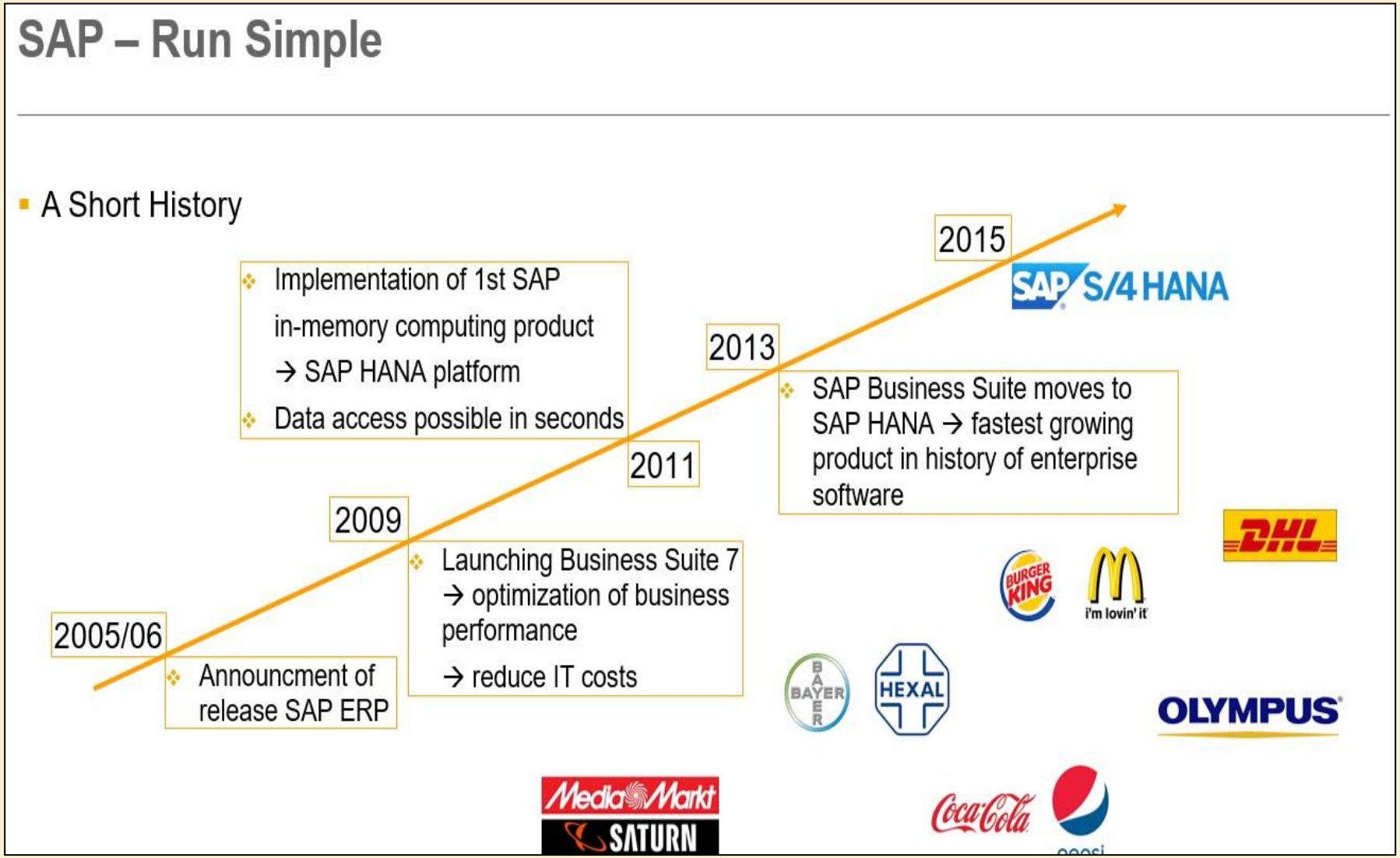


Figure 3: SAP – Run Simple. Source: Introduction to SAP S4HANA @2023 SAP SE/SAP UCC Magdeburg. All rights Reserved.

Development of ERP Systems (Cont.)

- SAP S/4HANA ...

... is the next generation Business Suite

... is the biggest innovation since SAP R/3

... connects people, business networks and devices

... works in real time

... represents efficiency, simplicity and innovation

... master data is managed centrally, for example partners, customers and vendors

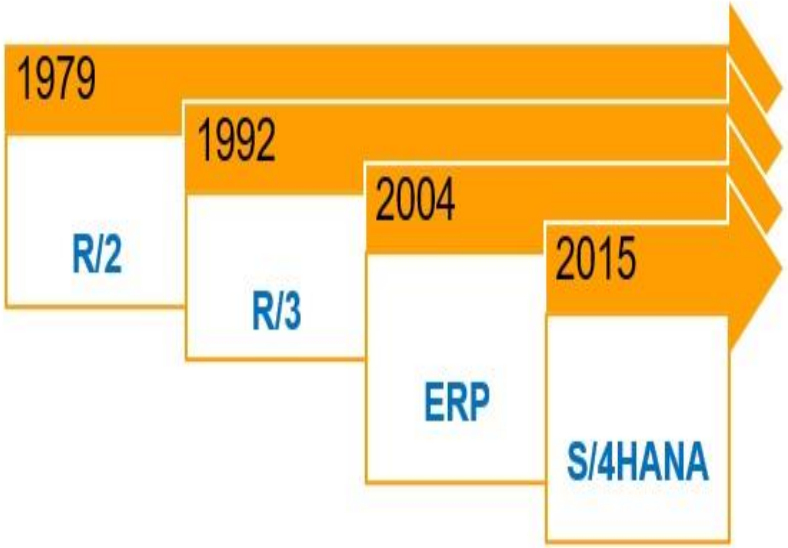


Figure 4: SAP S/4HANA. Source: Introduction to SAP S4HANA @2023 SAP SE/SAP UCC Magdeburg. All rights Reserved.

Development of ERP Systems (Cont.)

Enterprise Resource Planning (SAP ERP)

- Enables a company to support and optimize its business processes
- Helps the organization run smoothly
- Real-time environment, Scalable and flexible
- Collections of logically related transactions within identifiable business functions

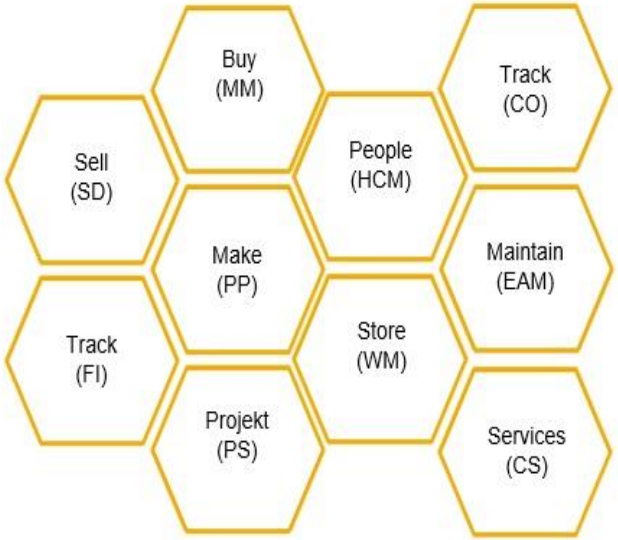
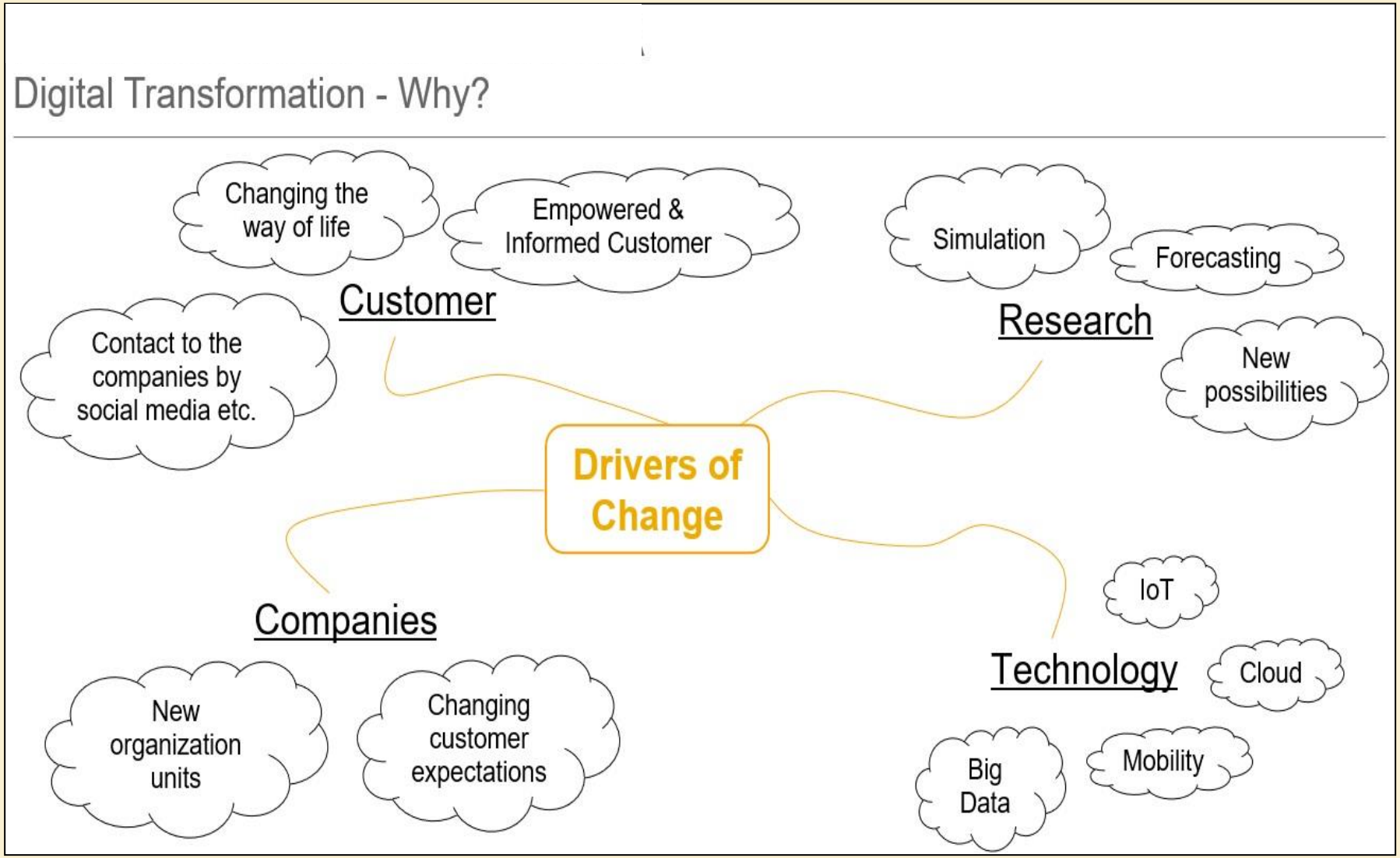


Figure 5: SAP ERP. Introduction to SAP S4HANA @2023 SAP SE/SAP UCC Magdeburg. All rights Reserved.

Development of ERP Systems (Cont.)



Source: Introduction to SAP S4HANA @2023 SAP SE/SAP UCC Magdeburg. All rights Reserved.

Development of ERP Systems (Cont.)

- S/4HANA is the digital core of the company that enables digital transformation.

Source: Introduction to SAP S4HANA @2023 SAP SE/SAP UCC Magdeburg. All rights Reserved.

In conclusion

- What is ERP?
- Example of Major ERP modules.
- Benefits of ERP.
- Challenges of ERP.
- Causes of ERP failures.
- Development of ERP Systems.

2. Business Function vs Business Process

Functional Areas and Business Processes

- To understand ERP, we must understand how a business works.
  - ❖ Functional areas of operation.
  - ❖ Business processes.

Functional Areas of Operation

- Most companies have four main functional areas of operation:
  - ❖ Marketing and Sales (M/S), Supply Chain Management (SCM), Accounting and Finance (A/F), and Human Resources (HR).
- **Business functions**: Activities specific to a functional area of operation.

Functional Areas of Operation (Cont.)

Functional area of operation	Marketing and Sales	Supply Chain Management	Accounting and Finance	Human Resources
Business functions	Marketing a product	Purchasing goods and raw materials	Financial accounting of payments from customers and to suppliers	Recruiting and hiring
	Taking sales orders	Receiving goods and raw materials	Cost allocation and control	Training
	Customer support	Transportation and logistics	Planning and budgeting	Payroll
	Customer relationship management	Scheduling production runs	Cash-flow management	Benefits
	Sales forecasting	Manufacturing goods		Government compliance
	Advertising	Plant maintenance		

Figure 6: Examples of functional areas of operation and their business functions.

Functional Areas of Operation (Cont.)

- Functional areas are interdependent.
  - ❖ Each requires data from the others.
- Better integration of functional areas leads to improvements in communication, workflow, and success of company.
- **Information system (IS)**: Computers, people, procedures, and software that store, organize, and deliver information.

Business Processes

- Collection of activities that takes one or more kinds of input and creates an output that is of value to customer.
  - ❖ Customer can be traditional external customer or internal customer.
- Thinking in terms of business processes helps managers to look at their organization from the customer's perspective.

Business Processes (Cont.)

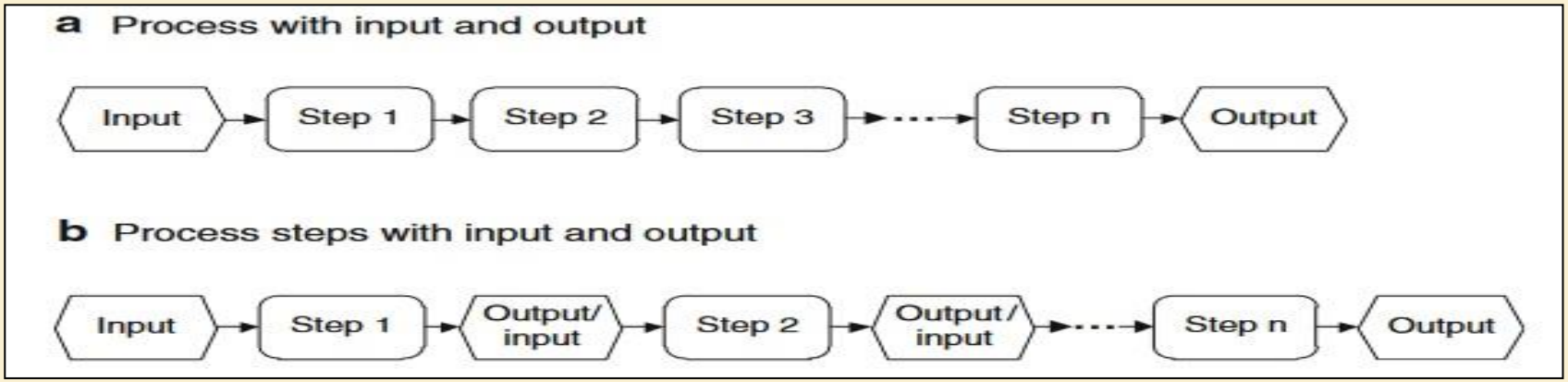


Figure 7: Schematic view of a business process.

Input	Functional area responsible for input	Process	Output
Request to purchase smartphone	Marketing and Sales	Sales order	Order is generated
Financial help for purchase	Accounting and Finance	Arranging financing in-house	Customer finances through the smartphone company
Fulfillment of order	Supply Chain Management	Shipping and delivery	Customer receives smartphone
Technical support	Marketing and Sales	24-hour help line available	Customer's technical query is resolved

Figure 8: Sample business processes related to the sale of a personal smartphone.

Business Processes (Cont.)

- Businesses must always consider customer's viewpoint in any transaction.
- Successful customer interaction.
  - ❖ Customer (either internal or external) is not required to interact with each business function involved in the process.
- Successful business managers view business operations from the perspective of a satisfied customer.

Business Processes (Cont.)

- Sharing data effectively and efficiently between and within functional areas leads to more efficient business processes.

- **Integrated information systems**: Systems in which functional areas share data.

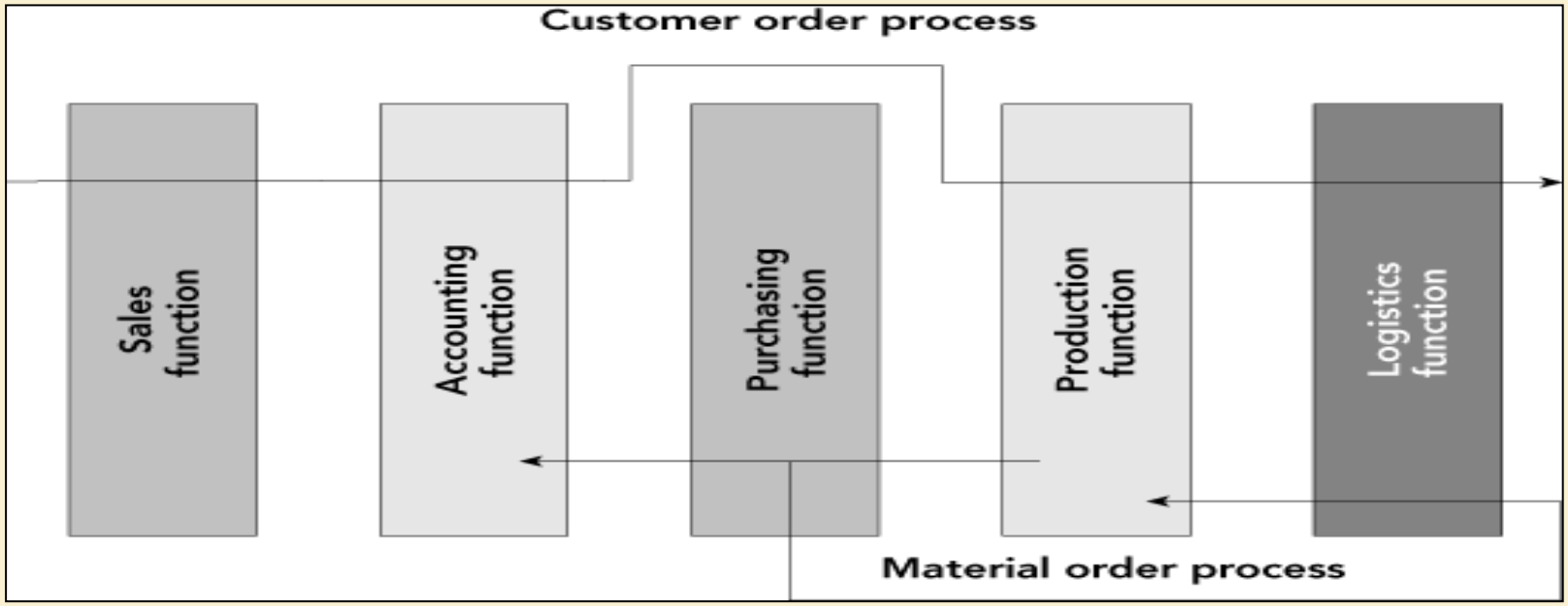


Figure 9: A process view of business.

Business Processes (Cont.)

- Businesses take inputs (resources) and transform these inputs into goods and services for customers.
  - ❖ Inputs: Material, people, equipment.
- Managing inputs and business processes effectively requires accurate and up-to-date information.

In conclusion

- Functional Areas and Business Processes.
- Functional Areas of Operation.
- Business Processes.

3. Client/Server Technology, 2-Tier, 3-Tier, n-Tier

Client/Server Technology

- The Client/Server computing has three components in it namely:
  - ❖ Client.
  - ❖ Server.
  - ❖ Network.
- Basic Client Server model:

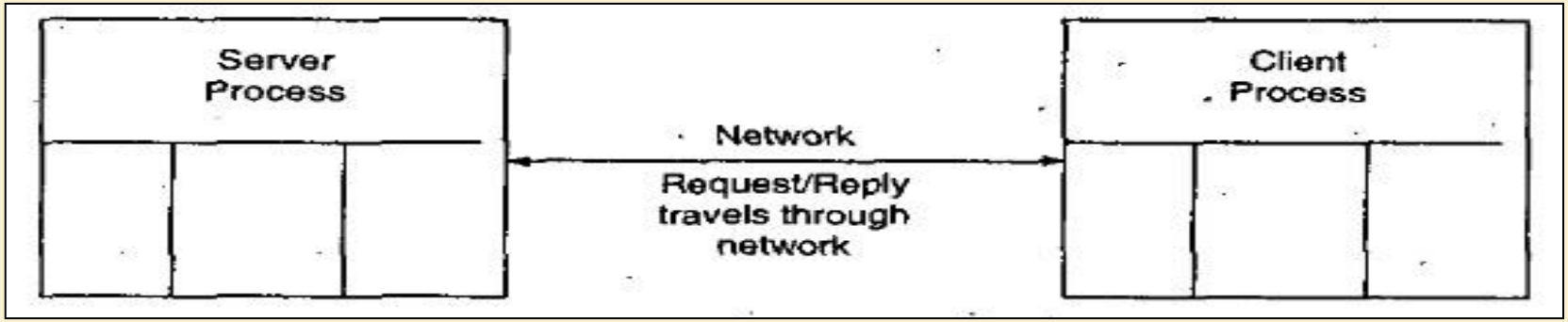


Figure 10: Basic Client Server Model. Source: Client Server Technology, Self Learning Material, SWAMI VIVEKANAND SUBHARTI UNIVERSITY, 2017.

Client/Server Technology (Cont.)

- *Server:* This is one that manages data or information in database. It also takes the role 'of coordination or assigning process when multiple clients functions which is very common in client/server computing technology.
- *Client:* This is the application or in other words called as front end which the user uses to interact. This is the interface through which the user provides data to database or retrieve data from database.
- *Network: A* link must be there to connect client and server for the application to function as a whole. This is done by network. This acts as a communication media between client and server.

Client/Server Technology (Cont.)

- Client Server model

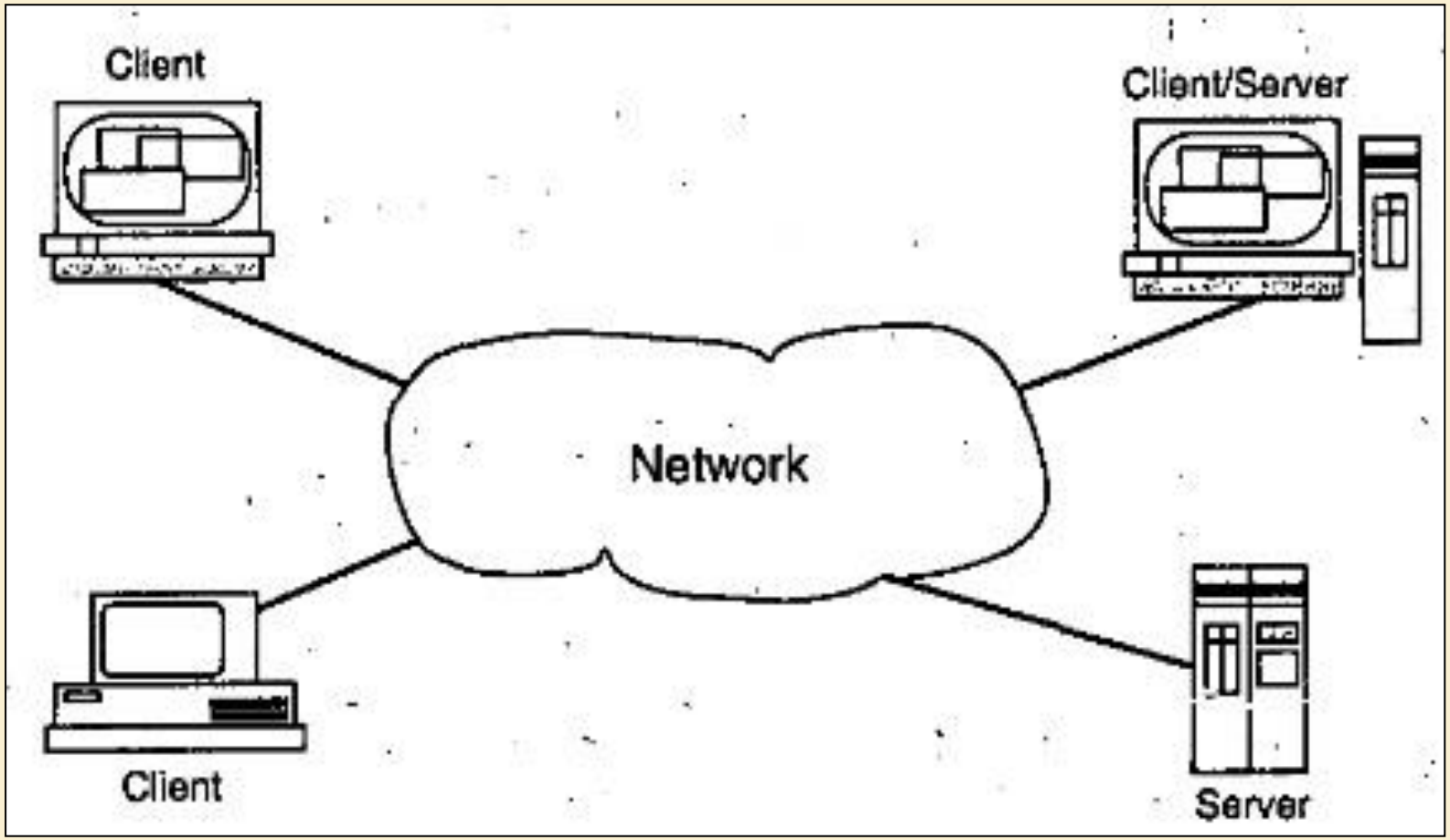


Figure 11: Client Server Model. Source: Client Server Technology, Self Learning Material, SWAMI VIVEKANAND SUBHARTI UNIVERSITY, 2017.

Client/Server Technology (Cont.)

- *Advantages:*
  - ❖ The type of design one can choose for a system depends on the needs of each component of a client-server computing environment.
  - ❖ An organization can make a cost-effective system by purchasing systems as per its needs.
    - ➢ Cost Effective; Increased Performance; Ease of effort and maintenance.
- *Disadvantages:*
  - ❖ Costs of clients' maintenance include users, developers, administrators to maintain the database on the server side, and so on.

Components of Client Server Model

- Client Server architecture is based on hardware and software components that interact to client form a server system. The system includes mainly three components.
  - ❖ Hardware (client and server).
  - ❖ Software (which make hardware operational).
  - ❖ Communication middleware, (associated with a network which are used to link the hardware and software).

Classification of Client Server Systems

- There are three types of client-server systems in existence as follows
  - ❖ Two-tier.
  - ❖ Three-tier.
  - ❖ N-Tier.

Source: https://www.guru99.com/n-tier-architecture-system-concepts-tips.html

Two-tier Client Server Model

- The 2-tier architecture is used to describe client-server systems where the client requests resources and the server responds directly to the request using its resources. This means that the server does not call on another application to provide part of the service.

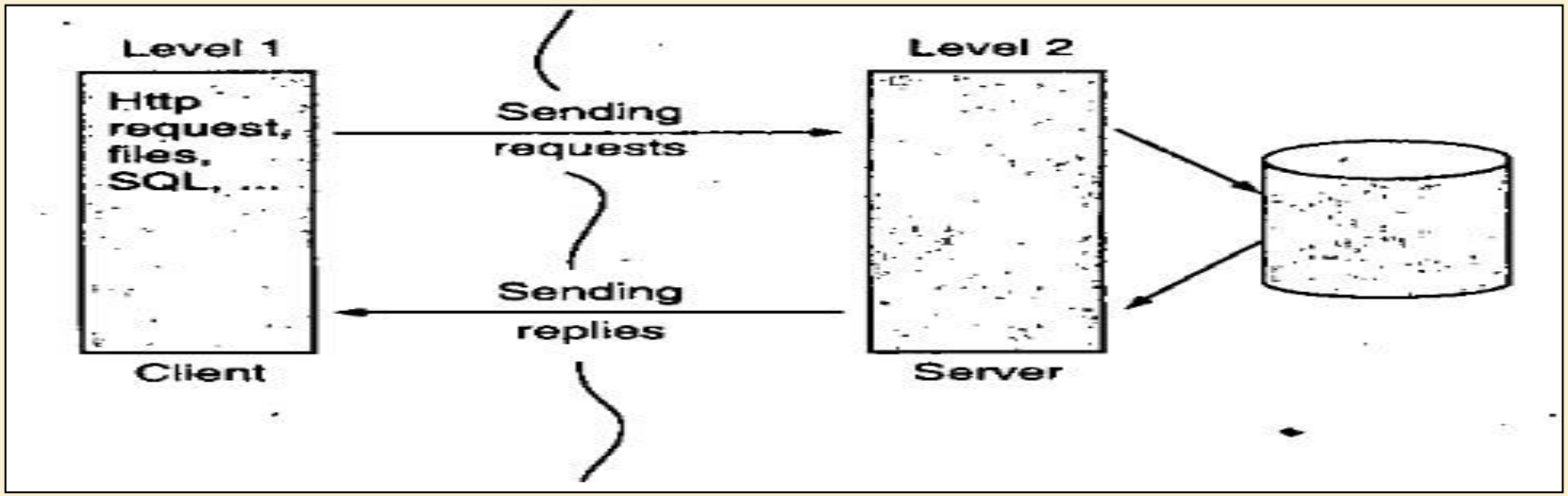


Figure 12: Two-tier Client Server Model. Source: Client Server Technology, Self Learning Material, SWAMI VIVEKANAND SUBHARTI UNIVERSITY, 2017.

Three-tier Client Server Model

- In 3-tier architecture, there is an intermediary level, meaning the architecture is generally split up between:
  - A client, i.e., the computer that requests the resources, is equipped with a user interface (usually a web browser) for presentation purposes.
  - The application server (also called middleware), whose task it is to provide the requested resources, calls on another server.
  - The data server provides the application server with the data it requires.

Three-tier Client Server Model (Cont.)

- In three-tier architecture application responsibilities are divided into three logical categories (in other words, the business system must provide three types of main services).
  - ❖ Presentation (GUI) or user services.
  - ❖ Application services or business rules.
  - ❖ Database services or data server.

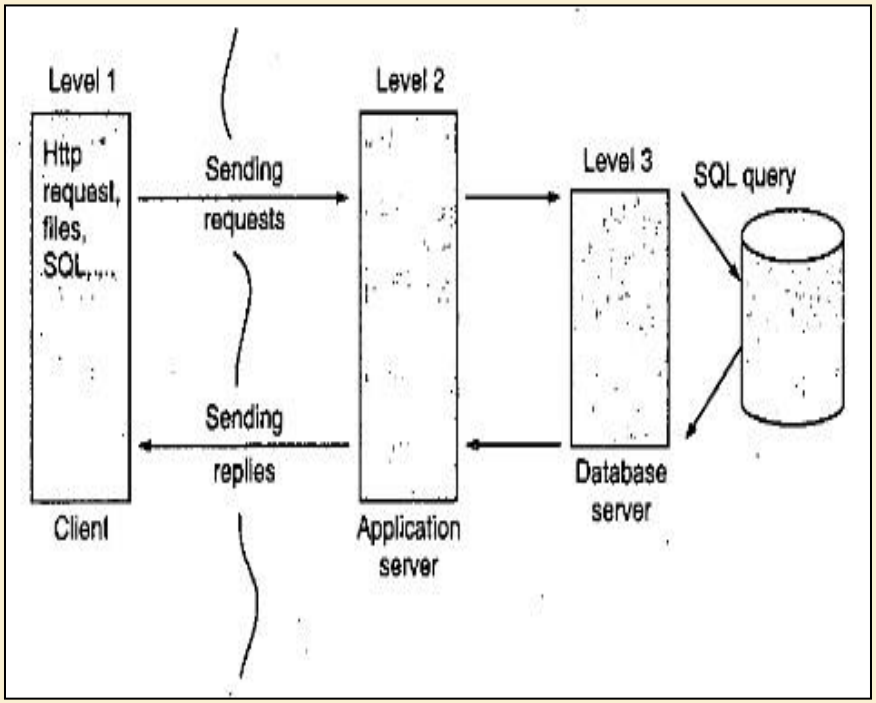


Figure 12: Three-tier Client Server Model. Source: Client Server Technology, Self Learning Material, SWAMI VIVEKANAN SUBHARTI UNIVERSITY, 2017.

Three-tier Client Server Model (Cont.)

- *Presentation (GUI)/user services*: Maintain the graphical user interface and generate what users see on the monitor.
  - ❖ Screen formatting; windows management; input editing; what-if analysis.
- *Application services/business rules*: Include executing applications and controlling program flow.
  - ❖ Domain and range validation; data dependency validation; request/response architecture of the inter-process communication level.
- *Database services/data servers*: Refer to the management of underlying databases.
  - ❖ Data access, management, and security; SQL parsing.

Comparing Both Types of Architecture

- 2-tier architecture is a client-server architecture where the server is flexible, i.e. it is capable of directly responding to all of the client's resource requests.
- In 3-tier architecture, however, the server-level applications are remote from one another, i.e. each server is specialized for a certain task (for example: web server or database server).
  - ❖ A greater degree of flexibility.
  - ❖ Increased security, as security can be defined for each service, and at each level.
  - ❖ Increased performance as tasks are shared between servers.

Multi-Tiered or N-tier Client Server Model

- In a 3-tier architecture, each server (tiers 2 and 3) performs a specialized task (a service). A server can therefore use services from other servers to provide its own service. As a result, 3-tier architecture is potentially an n-tiered architecture.
- Multi-Tiered /
  N-tier Client Server Model

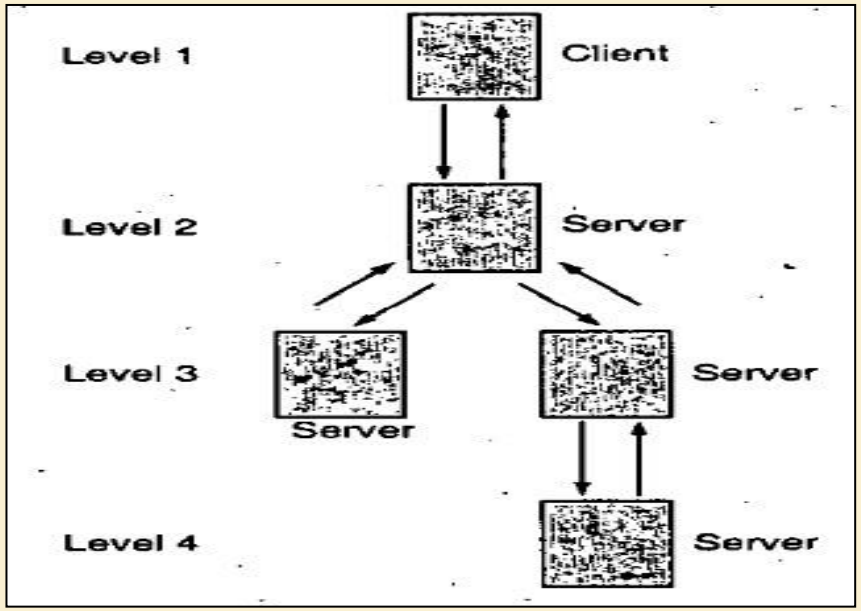


Figure 13: N-tier Client Server Model. Source: Client Server Technology, Self Learning Material, SWAMI VIVEKANAND SUBHARTI UNIVERSITY, 2017.

Architecture of Enterprise Systems

- The architecture of an enterprise system refers to the technical structure of the software. The three-tier client-server architecture separates these layers into three separate systems.

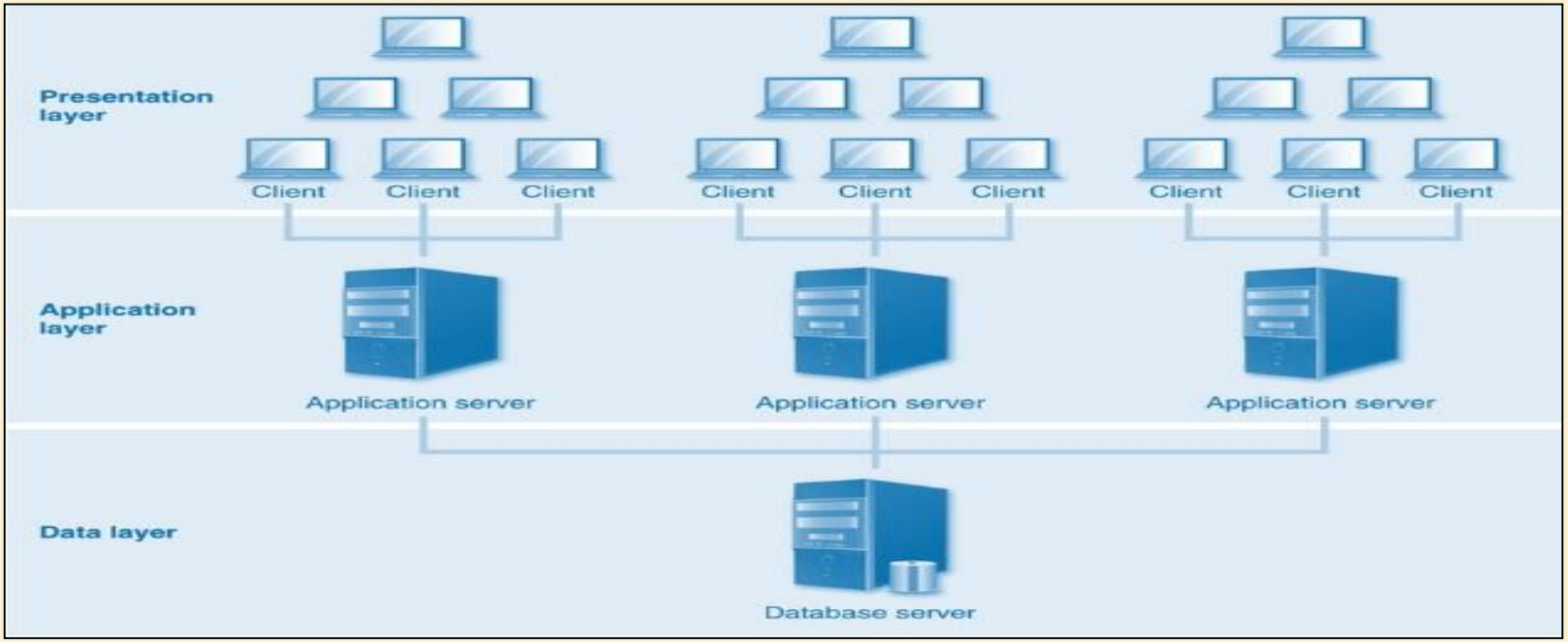


Figure 14: Architecture of Enterprise Systems.

Architecture of Enterprise Systems (Cont.)

- *The presentation layer* is responsible for the communication between the user and the system. It provides a graphical user interface, accepts and interprets user input and requests application programs accordingly.
- *The application layer* contains the business-oriented ERP functionality, decoupling the application logic from the user interface and the database.
- *The database layer* is in charge of the data the application layer works with. It stores an manages these data, usually with the help of a database server.

In conclusion

- Client/Server Technology.
- Components of Client Server Model.
- Classification of Client Server Systems.
- Two-tier Client Server Model.
- Three-tier Client Server Model.
- Multi-Tiered or N-tier Client Server Model.
- Architecture of Enterprise Systems.

4. Example of ERP – SAP

The Evolution of SAP ERP

- The best-known ERP system for years has been that of SAP AG, a German software company based in Walldorf, Baden. The name of the system, however, has changed several times over the years.

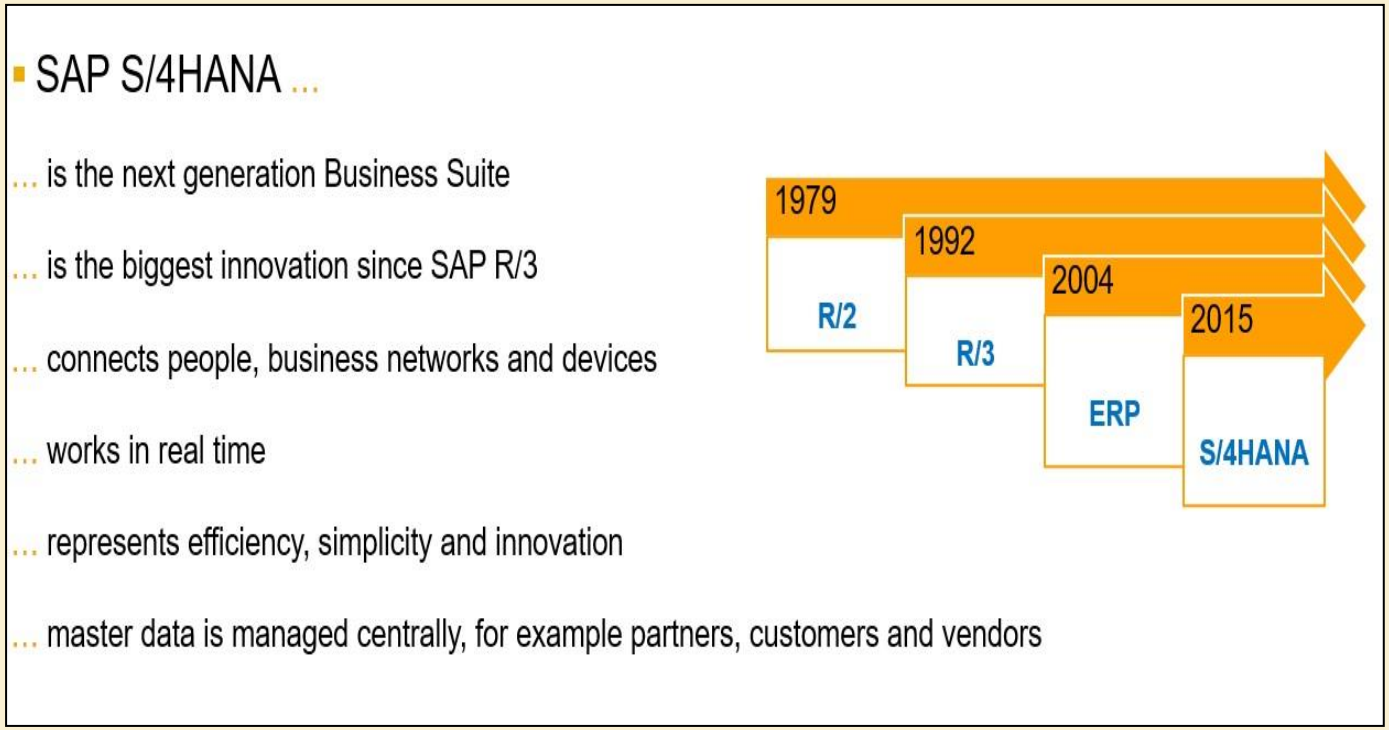


Figure 15: SAP S/4HANA. Source: Introduction to SAP S4HANA @2023 SAP SE/SAP UCC Magdeburg. All rights Reserved.

Solution Map of SAP ERP

- The "*solution map*," also called the "*business map*," is a diagram that summarizes the most important areas supported by SAP ERP.
- It provides an overview of business functions and processes.
- According to SAP, it should help to visualize, plan, and implement a consistent, integrated, and comprehensive business solution.

Solution Map of SAP ERP (Cont.)

- The SAP ERP solution map

Functionality of SAP ERP

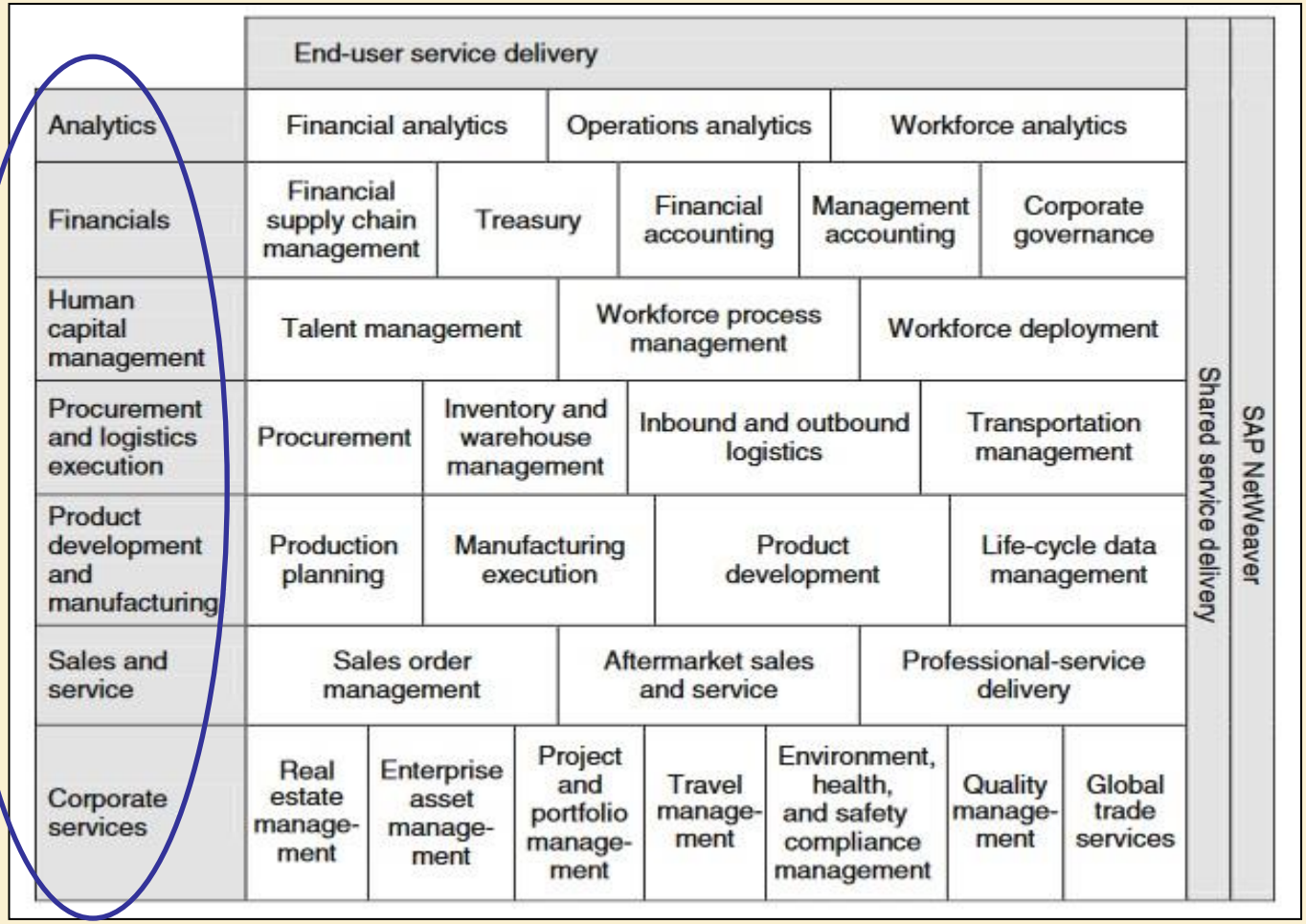


Figure 16: The SAP ERP solution map.

SAP S/4HANA Architecture

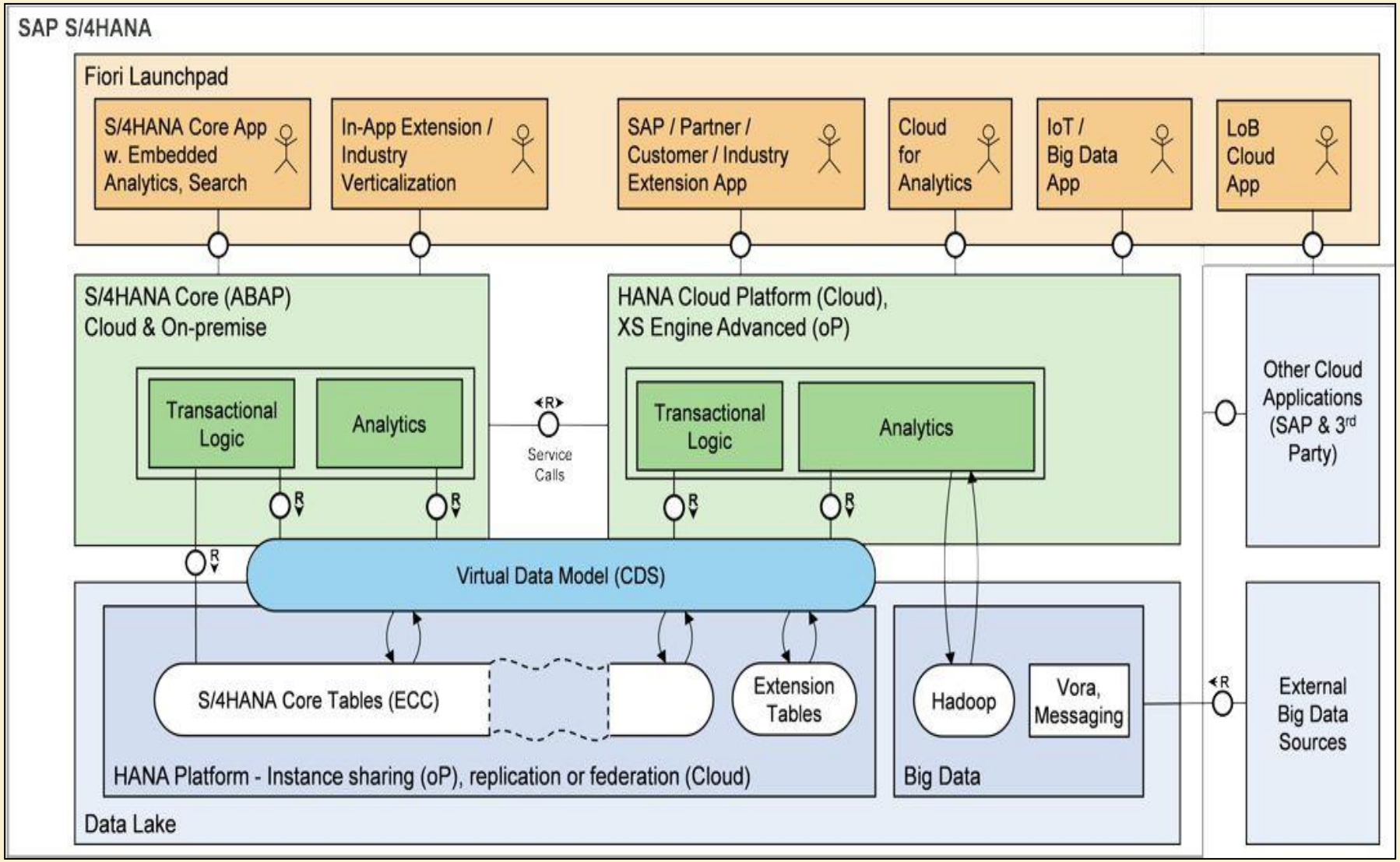
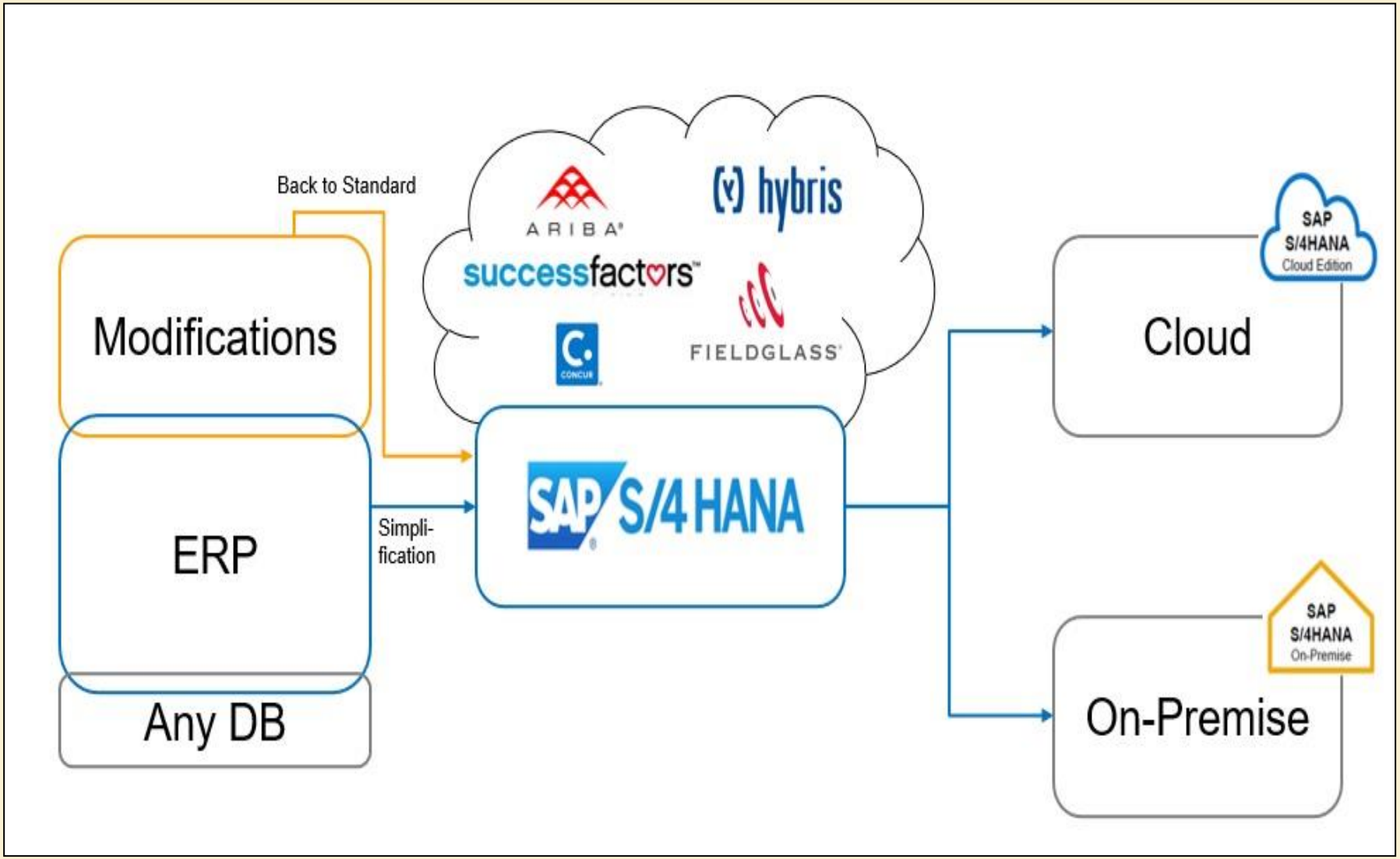


Figure 17: SAP S/4HANA Architecture. Source: Introduction to SAP S4HANA @2023 SAP SE/SAP UCC Magdeburg. All rights Reserved.

SAP S/4HANA Platform



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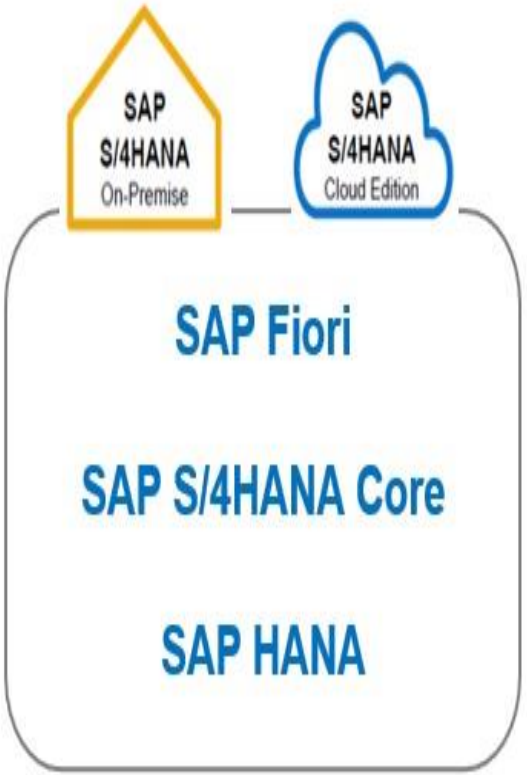
SAP S/4HANA Platform (Cont.)

On-Premise vs. Cloud

- SAP S/4HANA ...

... Choose between Cloud and On-Premises

- Cloud and On-Premise can be used hybridly without losing the company´s integration
- Possible migration of SAP Business Suite customers



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SAP S/4HANA Platform (Cont.)

On-Premise vs. Cloud

SAP S/4HANA Cloud Edition

Subscription Licensing

→ Deployment in the private cloud, maintained by SAP

- SAP provides system and controls maintenance
- Automatic participation in quarterly innovation upgrades
- In-App extensibility with limited ABAP
- Current release cycles
- SAP ERP embedded

SAP S/4HANA On-Premise

Traditional licensing

→ Traditional licensing with customer control of deployment and maintenance

- Private control of deployment and maintenance
- Hardware at companies location
- Privately controlled data
- Fewer release cycles
- Individual requirements possible
- Traditional ABAP extensibility up to core modification

Source: Introduction to SAP S4HANA @2023 SAP SE/SAP UCC Magdeburg. All rights Reserved.

SAP S/4HANA Platform (Cont.)

- SAP S/4HANA ...

... works with in-memory → **In-Memory Database**
- Column store
- In-memory processing
- Compression
- Parallel processing

... has a new and modern design → **Fiori**
- Hardware independence
- Mobile devices

... works with new technology → **OLAP+OLTP together**
- Real-time analytics on current data
- Consolidated forecasts including actual and planned cash data

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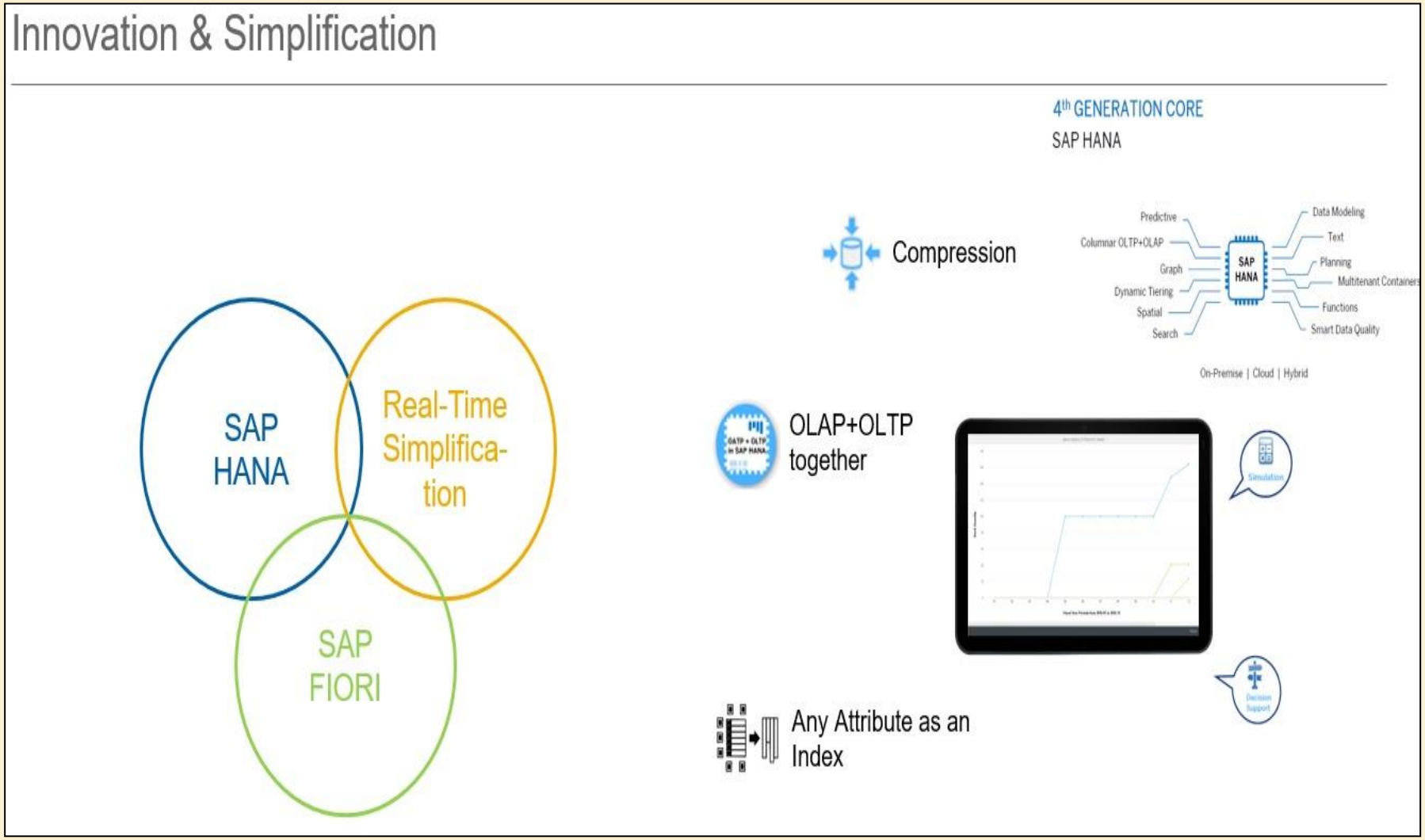
Innovation and Simplification



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SAP HANA

SAP HANA

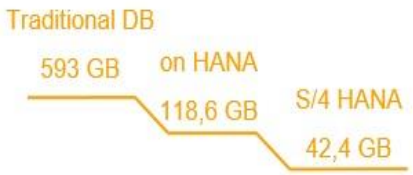
- SAP HANA is an in-memory database that allows you to process data very fast
- But SAP HANA can do way more with its different processing engines

Source: Introduction to SAP S4HANA @2023 SAP SE/SAP UCC Magdeburg. All rights Reserved.

Real Time Simplification

Real Time Simplification

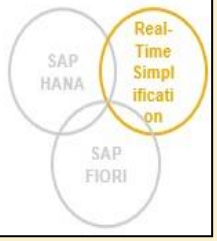
- Example of data compression



A typical booking transaction addressed 15 tables.
→ Now it only impacts 4 tables.

		Table sizes of customer system	
Table	Description	ERP Financials in Suite on HANA	Simple Finance on HANA
BKPF	Accounting documents	1.2 GB	1.2 GB
BSEG	Accounting document line items	4.5 GB	** 5.0 GB
BSID	Open items Accounts Receivable	0.1 GB	
BSIK	Open items Accounts Payable	0.0 GB	
BSIS	Open items General Ledger	* 23.2 GB	
BSAD	Cleared items Accounts Receivable	3.3 GB	
BSAK	Cleared items Accounts Payable	2.1 GB	
BSAS	Cleared items General Ledger	* 5.5 GB	
KNC1	Totals Accounts Receivable	0.1 GB	
LFC1	Totals Accounts Payable	0.0 GB	
GLT0	Totals General Ledger	0.0 GB	
	Total (of above tables)	40.0 GB	6.2 GB

Traditional Database Architecture	S/4HANA Data Model Principles
• Normalized data modeling third normal form • Avoidance of unwanted redundancies • Avoidance of inconsistencies and anomalies <u>Disadvantages</u> – Frequent use of redundant data to increase the performance of e.g. data aggregation – Higher effort to update redundant data	• Storage of data in denormalized form • Single source of data • No longer need for redundant data storage for tasks such as aggregation • Processing of aggregation and analytics on the fly • Check for inconsistencies and anomalies due to denormalization no longer a critical task



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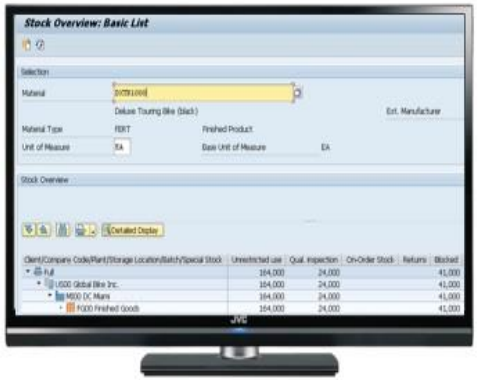
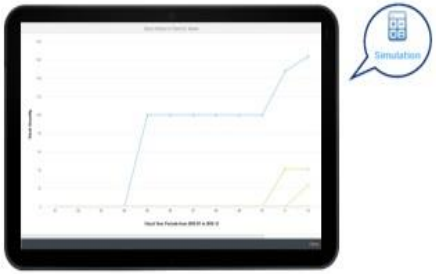
SAP FIORI

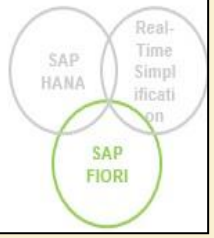
SAP FIORI

- SAP S/4HANA provides SAP Fiori user interface and experience



SAP S/4 HANA
- Independent of hardware
- Real-time





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SAP FIORI (Cont.)

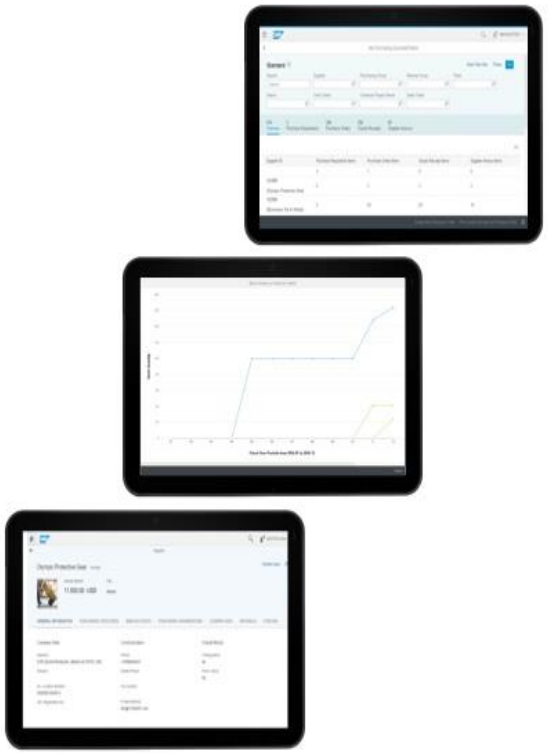
SAP FIORI

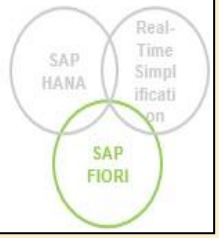
- SAP Fiori offers three application types:

- Transactional Apps:
Access to tasks like create, change or display process with guided navigation

- Analytical Apps:
Visual overview about business data

- Factsheet:
View essential information about objects and contextual navigation between related objects





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Data Types in ERP Systems

SAP S/4HANA
Data Types in ERP Systems

Organizational Data	Master Data	Transaction Data
Company Code	Person	Purchase Order
Plant	Material	Invoice
Storage Location	Customer	Quotation
Distribution Channel	Vendor	Sales Order
Purchasing Organization	Work Centre	Transportation Order
…	…	…

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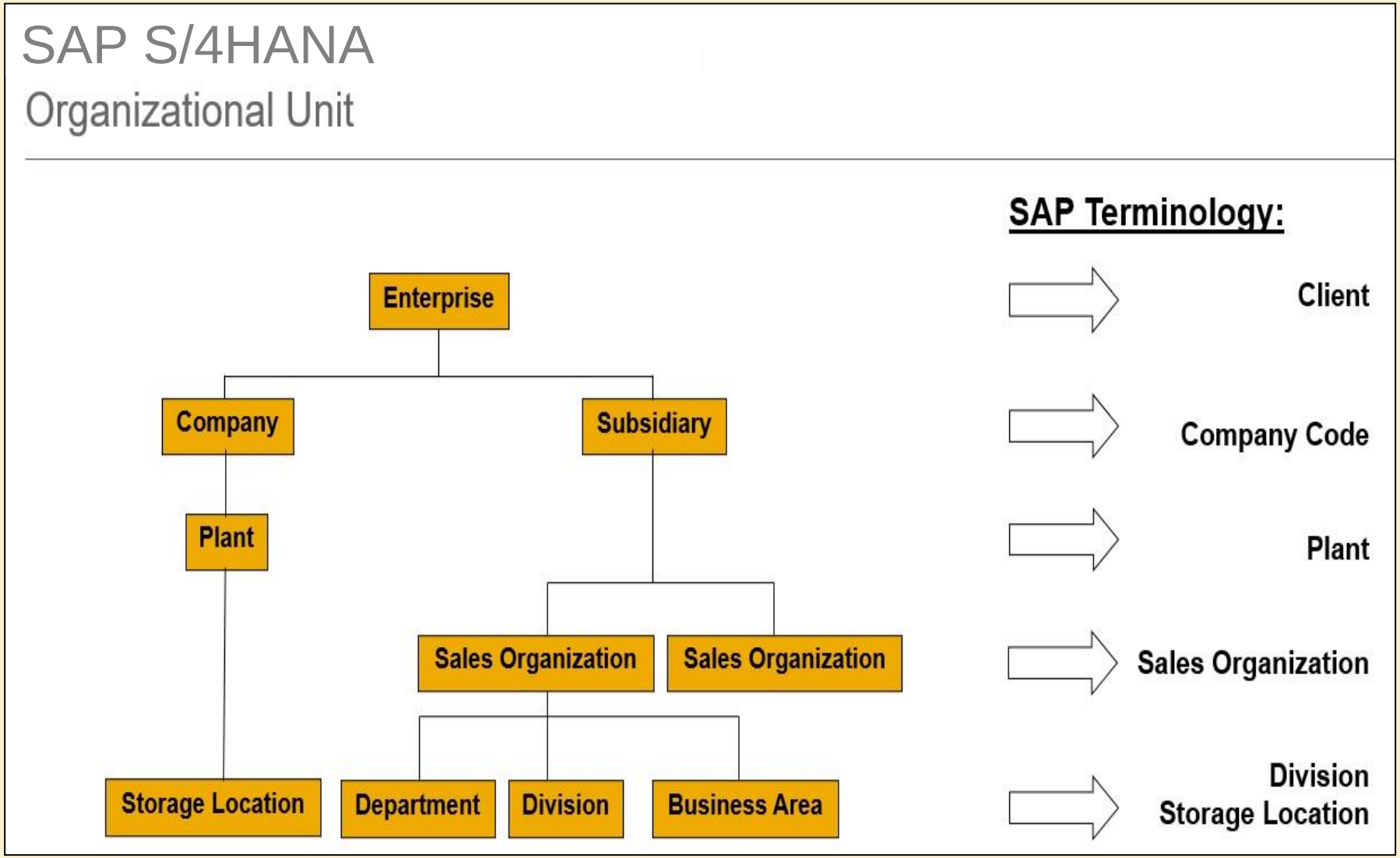
Data Types in ERP Systems (Cont.)



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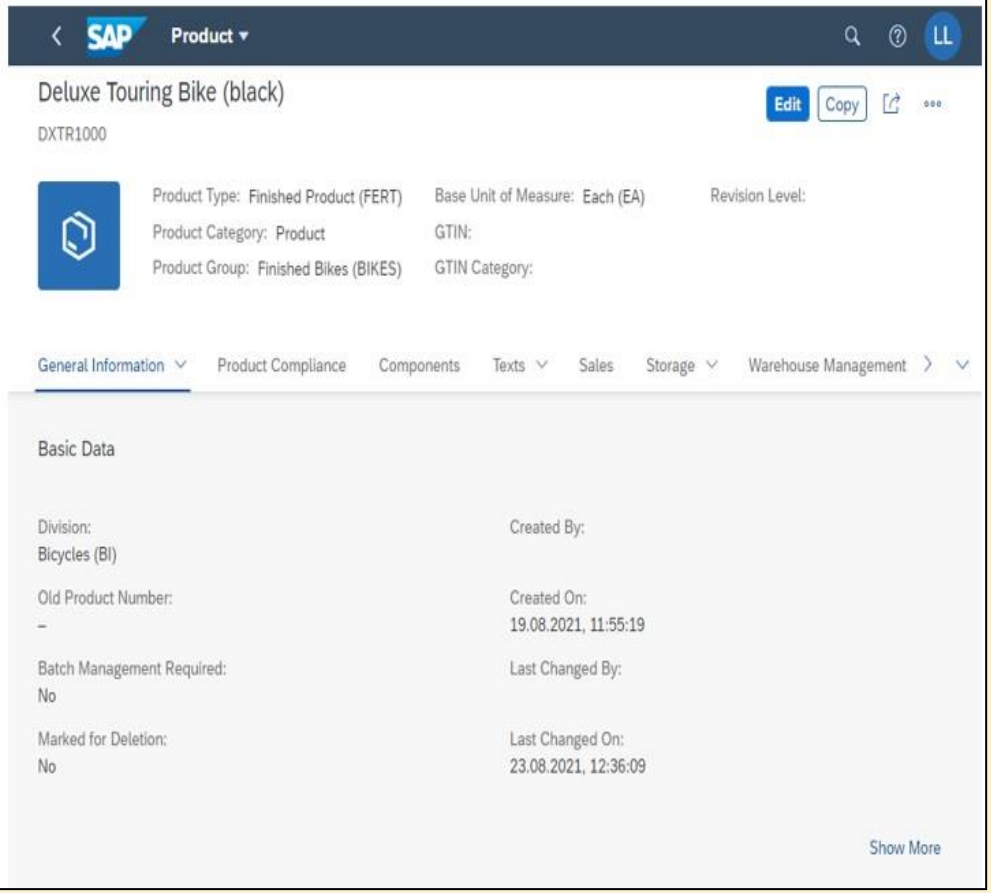
Data Types in ERP Systems (Cont.)

SAP S/4HANA
Master Data

- Stored for a long time and seldom changed
- Represent logically grouped data like:
  - Customer Master
  - Material Master
  - Vendor Master
  - General Ledger accounts



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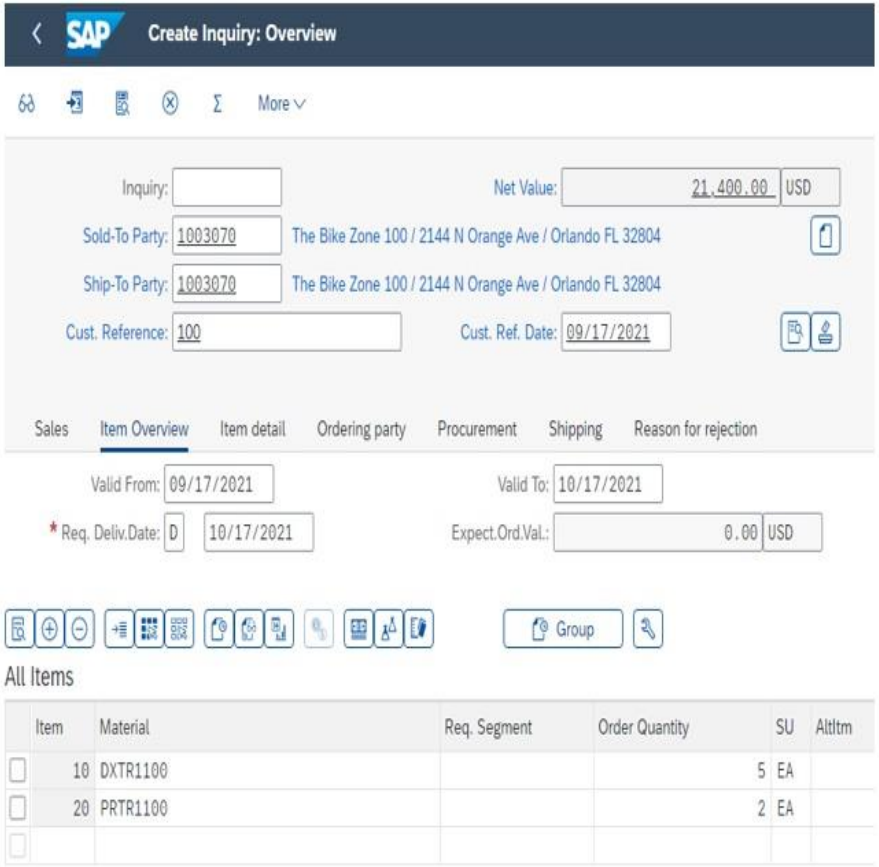
Data Types in ERP Systems (Cont.)

SAP S/4HANA
Transaction Data

- Transaction data is the system record of business event.
- Depending on the business event, different master data and organizational data will be referenced
- For example, during a sales order business event, the following data is stored
  - Organizational level: client, company code, sales organization
  - Master data: customer, material, pricing (condition)
  - Situational data: date, time, person, amount



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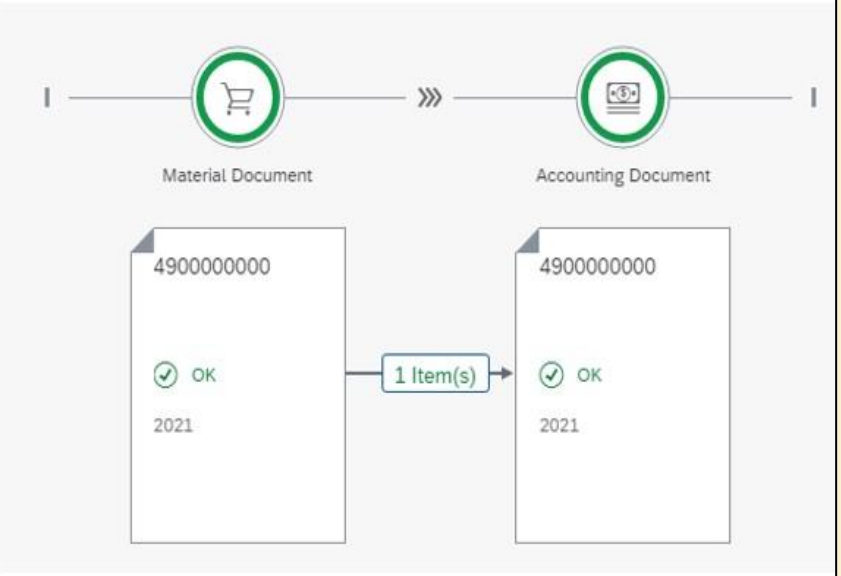
Data Types in ERP Systems (Cont.)

SAP S/4HANA
Documents

- Transactions are data sets that are generated if a business transaction was executed.
- Is a record of the business transaction
- Includes all relevant predefined information from the master data and organizational entities
- Example:
  - Sales Document
  - Purchasing Document
  - Material Document
  - Accounting Document

Document Flow

- The document flow as well as the order status allow the setting of the status at any point in time
- SAP revises the status every time a change in a document takes place



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SAP S/4HANA Application Modules

- SD: Sales & Distribution.
- FI: Financial Accounting.
- PS: Project Management.
- CS: Customer Service.
- MM: Material Management.
- CO: Controlling.

- HCM: Human Capital Management.
- PP: Production Planning & Execution.
- EAM: Enterprise Asset Management.
- WM: Warehouse Management.

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SAP S/4HANA Application Modules (Cont.)

- *Sales and Distribution (SD)*
  - ❖ SD is a core functional module in SAP ERP Central Component (ECC) that allows organizations to store and manage customer- and product-related data. Organizations use this data to manage all of the sales ordering, shipping, billing, and invoicing of their goods and services.
- *Project Management (PS)*
  - ❖ PS is an integrated project management tool used for planning and managing projects. This SAP module supports the entire project lifecycle from planning to invoicing.
-

SAP S/4HANA Application Modules (Cont.)

- *Financial Accounting (FI)*
  - ❖ The FI module (financials) offers a multitude of functions covering the general ledger, accounts receivable, and accounts payable, as well as other types of accounting (assets, inventory, tax, bank-related accounting, and more).
- *Customer Service (CS)*
  - ❖ Customer Service Module provides your customer service agents (CSAs) with easy and fast access to the information needed to understand and quickly resolve customers' problems.

SAP S/4HANA Application Modules (Cont.)

- *Material Management (MM)*
  - ❖ MM is an SAP ERP component that helps organizations with material management, inventory management, and warehouse management in the supply chain process.
  - ❖ It is a part of SAP ECC's logistics functions, which consist of several components and sub-components. The most prominent and widely used components are master data, purchasing, and inventory.

SAP S/4HANA Application Modules (Cont.)

- *Controlling (CO)*
  - ❖ CO provides you with information for management decision-making. It facilitates the coordination, monitoring, and optimization of all processes in an organization.
- *Human Capital Management (HCM)*
  - ❖ Human capital management software refers to the systems and solutions used to accomplish and optimize those tasks and support organizational goals and is often referred to as human resources management systems (HRMS).

SAP S/4HANA Application Modules (Cont.)

- *Production Planning & Execution (PP)*
  - ❖ PP is a module that focuses on managing and optimizing the production processes within an organization, such as production planning, scheduling, and execution.
- *Enterprise Asset Management (EAM)*
  - ❖ EAM helps organizations to plan, optimize, execute, and track the necessary activities, priorities, skills, materials, tools, and information associated with an asset.

SAP S/4HANA Application Modules (Cont.)

- *Warehouse Management (WM)*
  - ❖ WM enables you to perform physical inventory and cycle counting procedures to maintain inventory accuracy and ensure compliance with auditing and regulatory requirements.
  - ❖ The system supports various physical inventory methods, such as periodic inventory, continuous inventory, and ad hoc inventory counting.

In conclusion

- The Evolution of SAP ERP.
- Solution Map of SAP ERP.
- SAP S/4HANA Architecture.
- SAP S/4HANA Platform.
- Innovation and Simplification.
- SAP HANA.
- Real Time Simplification.
- SAP FIORI.
- Data Types in ERP Systems.
- SAP S/4HANA Application Modules.

5. SAP S/4HANA (using Global Bike)

THANK YOU
FOR YOUR ATTENTION

Q&A